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Ramanujan's Master formula

if $f(x) = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} \lambda(n)$ —— (I)

then Mellin Transform of $f(x)$ is given by

$$* \int_0^{\infty} x^{s-1} f(x)\, dx = \Gamma s \; \lambda(-s) \quad \text{—— (II)}$$

Eg:- let $a > 0$ & $f(x) = e^{-ax}$

then $\int_0^{\infty} x^{s-1} e^{-ax}\, dx = ?$

Here $f(x) = e^{-ax} = \sum_{n=0}^{\infty} \frac{a^n (-x)^n}{n!}$

So $\int_0^{\infty} x^{s-1} e^{-ax}\, dx = \sum \int_0^{\infty} (x^{s-1})\, a^n \frac{(-x)^n}{n!}\, dx$

Comparing with formula -

$\lambda(n) = a^n$

using Ramanujan's formula

$$\int_0^{\infty} x^{s-1} e^{-ax}\, dx = \Gamma s \; \lambda(-s) = \Gamma s \; a^{-s}$$

$$\boxed{= \frac{\Gamma s}{a^s}}$$

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$\int_0^\infty x^{s-1} e^{-ax} dx = \Gamma s \, \lambda(-s) = \frac{\Gamma s}{a^s}$

Justisify with Eular gamma fn,

Ex-2 $\int_0^\infty \frac{x^{s-1}}{(1+x)^a} dx$,

using binomial th. expansion of

$$(1+x)^{-a} = \sum_{n=0}^{\infty} \frac{\Gamma n+a}{\Gamma a} \frac{(-x)^n}{n!}$$

the Integral

$$\int_0^\infty \frac{x^{s-1}}{(1+x)^a} dx = \sum \int_0^\infty x^{s-1} \left\{ \frac{\Gamma n+a}{\Gamma a \; n!} (-x)^n \right\} dx$$

$$\lambda(n) = \frac{\Gamma n+a}{\Gamma a}$$

using formula

$$\int_0^\infty x^{s-1} f(x) dx = \Gamma s \, \lambda(-s) \quad \text{gives}$$

$$\boxed{= \Gamma s \frac{\Gamma a-s}{\Gamma a}} = \underbrace{\beta(s, a-s)}_{\text{Beta fn}}$$

Relation b/w gamma & zeta fⁿ

$$\Gamma s = \int_0^\infty x^{s-1} e^{-x}\, dx$$

let $x = nu$, $dx = n\,du$

$$\Gamma s = \int_0^\infty n^s u^{s-1} e^{-nu}\, du$$

or
$$\frac{\Gamma s}{n^s} = \int_0^\infty u^{s-1} e^{-nu}\, du$$

Summing for all ~~+ve~~ integer n

$$\Gamma s \underbrace{\sum_{n=1}^\infty \frac{1}{n^s}}_{\zeta(s)} = \sum_{n=1}^\infty \int_0^\infty u^{s-1} e^{-nu}\, du$$

or
$$\Gamma s\ \zeta(s) = \int_0^\infty \left(u^{s-1} \sum_{n=1}^\infty e^{-un}\right) du$$

apply G.P to sum $\sum_{n=1}^\infty e^{-un} = \frac{1}{e^u - 1}$

therefor

$$\boxed{\Gamma s\ \zeta(s) = \int_0^\infty \frac{u^{s-1}}{e^u - 1}\, du}$$

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$$\int_0^1 x^x\,dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)^{n+1}}$$ $\boxed{\approx 0.79}$

Sol:

$$\int_0^1 e^{\ln x^x}\,dx = \int_0^1 e^{x\ln x}\,dx$$

$$= \sum_{n=0}^{\infty} \int_0^1 \frac{(x\ln x)^n}{n!}\,dx$$

let $\ln x = -t$

$dx = -x\,dt$

$$\sum_{n=0}^{\infty} \int_{\infty}^{0} \frac{(e^{-t}(-t))^n}{n!}\,dt\,(-)e^{-t}$$

$$= \sum_{n=0}^{\infty} \int_0^{\infty} \frac{e^{-(n+1)t}(t)^n(-1)^n}{n!}\,dt$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{\cancel{\Gamma(n+1)}}{\cancel{n!}\,(n+1)^{n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)^{n+1}}$$

Power - Series Expansion of a f^n

1) Taylor - Maclaurin Series

Suppose $f(x)$ has a power-series expansion at $x=a$ with radius of curvature $R > 0$

then -

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n = f(a) + \frac{f'(a)(x-a)}{1!} \cdots$$

if $a=0$ series called Maclaurin.

Ex) $f(x) = e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$$\frac{1}{R} = \lim_{n\to\infty} \left| \frac{C_{n+1} x^{n+1}}{C_n x^n} \right| = \frac{|x|}{n+1} = 0$$

$R = \infty$, expansion valid for all R.

if centre is a, ie $f(x) = e^{(x-a)}$

then

$$f(a) = \sum_{n=0}^{\infty} \frac{e^a (x-a)^n}{n!}, \quad R \to \infty$$

1) $f(x) = \sin ax = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} (a)^{2n+1}$

$$= \sum_{n=1}^{\infty} (-a)^n \sin\left(\frac{n\pi}{2}\right) \frac{(-x)^n}{n!}$$

therefor

$$\int_0^{\infty} x^{s-1} \sin ax \, dx = \sum_{n=1}^{\infty} \int_0^{\infty} x^{s-1} \underbrace{(-a)^n \sin \frac{n\pi}{2}}_{\lambda(n)} \frac{(-x)^n}{n!}$$

$$= \Gamma s \ \lambda(-s)$$

$$= \Gamma s \ \cancel{\cos} \ (-a)^{-s} \sin\left(-\frac{s\pi}{2}\right) = \Gamma s \ (a)^{-s} \sin \frac{s\pi}{2}$$

$$\boxed{\text{or } \int_0^{\infty} x^{s-1} \sin ax \, dx = \Gamma s \ (a)^{-s} \sin \frac{s\pi}{2}}$$

2) $f(x) = \cos ax = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{2n!} (a)^{2n}$

$$\text{or } \sum_{n=1}^{\infty} (-a)^n \cos\left(\frac{n\pi}{2}\right) \frac{(-x)^n}{n!}$$

therefor

$$\int_0^{\infty} x^{s-1} \sin ax \, dx = \text{Img.} \int_0^{\infty} x^{s-1} e^{iax} dx$$

$$= \sum \int_0^{\infty} x^{s-1} (ia)^n \frac{x^n}{n!}$$

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$$\int_0^\infty x^{s-1}\cos ax\,dx = \sum_{n=1}^{\infty}\int_0^\infty x^{s-1}\underbrace{(a)^n\cos\left(\frac{n\pi}{2}\right)}_{\lambda(n)}\frac{(-x)^n}{n!}$$

$$= \Gamma s\ \lambda(-s) = \Gamma s\ (a)^{-s}\cos\left(-\frac{s\pi}{2}\right)$$

$$= \Gamma s\ (-a)^{-s}\cos\left(\frac{s\pi}{2}\right)$$

or
$$\boxed{\int_0^\infty x^{s-1}\cos ax\,dx = \Gamma s\ (a)^{-s}\cos\left(\frac{s\pi}{2}\right)}$$

$s \in (-1, 1)$ $\boxed{s=1}$ for convergence of Integral.

3) $f(x) = \sin^2(ax)$

$$\sin^2 ax = \frac{1}{2}(1-\cos 2ax)$$

$$= \frac{1}{2}\left\{1-\sum_{n=0}^{\infty}(-1)^n\frac{(2ax)^{2n}}{2n!}\right\}$$

$$= \frac{1}{2}\left(1-\sum_{n=0}^{\infty}(-2a)^n\cos\left(\frac{n\pi}{2}\right)\frac{(-x)^n}{n!}\right)$$

$$= \frac{1}{2}\left(1-1+\sum_{n=1}^{\infty}(-2a)^n\cos\left(\frac{n\pi}{2}\right)\frac{(-x)^n}{n!}\right.$$

$$= \frac{1}{2}\sum_{n=1}^{\infty}(-2a)^n\cos\left(\frac{n\pi}{2}\right)\frac{(-x)^n}{n!}$$

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Therefore

$$\int_0^\infty x^{s-1} \sin^2 ax \, dx = \frac{-1}{2}\sum_{n=1}^{\infty} \int_0^\infty x^{s-1} \underbrace{(-2a)^n \cos\frac{n\pi}{2}}_{\lambda(n)} \frac{(-x)^n}{n!}$$

$$= \Gamma s \ \frac{-1}{2} \Gamma s \ \lambda(-s) = \frac{-1}{2}\Gamma s \ (-2a)^{-s} \cos\frac{-s\pi}{2}$$

$$= \Gamma s \ (-2)^{-s-1} (a)^{-s} \cos\frac{\pi s}{2}$$

or

$$\boxed{\int_0^\infty x^{s-1} \sin^2 ax \, ds = \Gamma s \ (-2)^{-s-1} (a)^{-s} \cos\frac{s\pi}{2}}$$

Note

$$\int_0^\infty x^{s-1}\cos^2 ax = ??$$

HW

$$\int_0^\infty x^{s-1} \sin^2 ax =$$

$$\int_0^\infty \frac{\sin x}{x} dx = \int_0^\infty \sum (-1)^n \frac{x^{2n}}{(2n+1)!} dx$$

$$x^2 = t \qquad dx = \frac{1}{2\sqrt{t}} dt$$

$$\frac{1}{2}\int_0^\infty t^{-1/2} \sum (-1)^n \frac{t^n}{(2n+1)!} \frac{n!}{n!} dt$$

$$\frac{1}{2}\Gamma\frac{1}{2} \frac{\Gamma\frac{1}{2}}{\Gamma 3/2} = \boxed{\frac{\pi}{2}}$$

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4) $f(x) = \cos^2 ax$

$= \frac{1 + \cos 2ax}{2}$

$= \frac{1}{2}\left\{ 1 + \sum_{n=0}^{\infty} \frac{\cancel{(-1)^n} (-2ax)^{2n}}{2n!} \right\}$

$= \frac{1}{2}\left\{ 1 + \sum_{n=0} (-2a)^n \cos\left(\frac{n\pi}{2}\right) \frac{(-x)^n}{n!} \right\}$

$= \frac{1}{2}\left\{ 2 + \sum_{n=1}^{\infty} (-2a)^n \cos\frac{n\pi}{2} \frac{(-x)^n}{n!} \right\}$

$= 1 + \frac{1}{2} \sum_{n=1}^{\infty} (-2a)^n \cos\frac{n\pi}{2} \frac{(-x)^n}{n!}$

Therefore

$\int_0^\infty x^{s-1} \cos^2 ax = \frac{1}{2} \sum_{n=1} \int_0^\infty x^{s-1} (2a)^n \cos\frac{n\pi}{2} \frac{(-x)^n}{n!} dx$

$\quad \lambda(n) = \frac{1}{\cdot}$

$+ \cancel{\int_0^\infty x^{s-1} dx}$

if $s<0$ $\quad -2 < Re(s) < 0$

then

$\frac{1}{2} \Gamma s \ \lambda(-s)$

$= \frac{1}{2} \Gamma s \ (-2a)^{-s} \cos\frac{-s\pi}{2}$

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(c) $\boxed{\int_0^\infty x^{s-1}\cos^2 ax\,dx = \Gamma s\,(2)^{-s-1}(-a)^{s}\cos\frac{\pi s}{2}}$

5) $\int_0^\infty \frac{1}{1+x^n}\,dx = \sum_{K=0}^{\infty}\int_0^\infty (-1)^K x^{nK}\,dx$

let $x^n = t$, $x = (t)^{1/n}$, $dx = \frac{1}{n}(t)^{1/n-1}\,dt$

$\sum_{K=0}^{\infty}\frac{1}{n}\int_0^\infty (-1)^K (t)^K \frac{dt}{\cancel{(t)^{\frac{1}{n}-1}}} (t)^{\frac{1}{n}-1}$

$= \sum_{K=0}^{\infty}\frac{1}{n}\int_0^\infty (t)^{\frac{1}{n}-1} \underbrace{(K!)}_{\lambda(n)} \frac{(-t)^K}{K!}$

$= \frac{1}{n}\Gamma\frac{1}{n}\,\lambda\left(-\frac{1}{n}\right)$

$= \frac{1}{n}\Gamma\frac{1}{n}\left(-\frac{1}{n}\right)!$

$= \frac{1}{n}\Gamma\frac{1}{n}\,\Gamma\left(1-\frac{1}{n}\right)$

$\left\{ \because \Gamma\frac{1}{n}\Gamma\left(1-\frac{1}{n}\right) = \frac{\pi}{\sin\frac{\pi}{n}} \right.$

$\boxed{\int_0^\infty \frac{1}{1+x^n}\,dx = \frac{1}{n}\frac{\pi}{\sin\frac{\pi}{n}}}$

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6) $\cancel{\int_0^\infty x}$ $f(x) = \sin(x^n)$

f $\sin(x^n) = \sum_{k=0}^{\infty} (-1)^k \frac{(x^n)^{2k+1}}{(2k+1)!}$

$\cancel{\sum \frac{(k\pi)}{2} \frac{(x^n)^k}{n!}}$

$\sum_{k=1}^{\infty} \sin\left(\frac{k\pi}{2}\right) \frac{(x^n)^{k}}{k!}$

Method – 1

$$I_n(s) = \int_0^\infty x^{s-1} \sin(x^n)\,dx$$

$$= \int_0^\infty x^{s-1} \sum_{k=0}^{\infty} (-1)^k \frac{(x^n)^{2k+1}}{(2k+1)!}\,dx$$

let $x^n = t \rightarrow x = (t)^{1/n}$

$dx = \frac{1}{n} (t)^{\frac{1}{n}-1}\,dt$

$$\int_0^\infty t^{\frac{s-1}{n}} \sum_k (-1)^k \frac{(t)^{2k+1}}{(2k+1)!}\,dt \left(\frac{1}{n}\right) t^{\frac{1}{n}-1}$$

$$\frac{1}{n}\int_0^\infty (t)^{\frac{s}{n}} \sum_{k=} (-1)^k \frac{(t)^{2k}}{(2k+1)!}\,dt$$

let $(t) = \sqrt{l}$

$dt = \frac{1}{2\sqrt{l}}\,dl$

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$$= \frac{1}{n}\int_0^{\infty} (l)^{\frac{s}{2n}} \sum_{k=}^{\infty} (-1)^k \frac{(l)^k}{(2k+1)!} \cdot \frac{1}{2\sqrt{l}}\, dl$$

$$= \frac{1}{2n}\int_0^{\infty} l^{\frac{s}{2n}-\frac{1}{2}} \sum_{k=0}^{\infty} \cancel{(-1)}\frac{1}{(2k+1)!} \frac{(-l)^k}{k!} \cdot (k!)\, dl$$

$$= \frac{1}{2n}\sum_{k=} l^{\frac{s}{2n}-\frac{1}{2}} \underbrace{\frac{k!}{(2k+1)!}}_{\lambda(k)} \frac{(-l)^k}{k!}\, dl$$

$$= \frac{1}{2n}\Gamma\left(\frac{s}{2n}+\frac{1}{2}\right) \lambda\left(-\left\{\frac{s}{2n}+\frac{1}{2}\right\}\right)$$

$$= \frac{1}{2n}\Gamma\left(\frac{s}{2n}+\frac{1}{2}\right) \frac{\left\{\left[-\left(\frac{s}{2n}+\frac{1}{2}\right)\right]!\right\}}{\left\{\left[-\left(\frac{s}{n}+1\right)\right]+1\right\}!}$$

$$= \frac{1}{2n}\Gamma\left(\frac{s}{2n}+\frac{1}{2}\right) \frac{\Gamma\left(1-\left(\frac{s}{2n}+\frac{1}{2}\right)\right)}{\Gamma\left(2-\left(\frac{s}{n}+1\right)\right)} \quad \left\{ \because \Gamma n\, \Gamma 1-n = \frac{\pi}{\sin n\pi} \right.$$

$$= \frac{\frac{1}{2n} \cancel{\sin} \left\{\frac{\pi}{\sin\left(\frac{s+n}{2n}\right)\pi}\right\}}{\Gamma\left(2-\left(\frac{s}{n}+1\right)\right)}$$

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$$= \frac{1}{2n} \frac{\pi}{\sin\left(\frac{s}{2n}\pi + \frac{n}{2n}\pi\right)}$$
$$\overline{\sqrt{1-\left(\frac{s}{n}\right)}}$$

$$\left\{ \sqrt{1-\left(\frac{s}{n}\right)} \quad \frac{1}{\sqrt{\frac{s}{n}}} \frac{\pi}{\sin\left(\frac{s}{n}+1\right)\pi} \right.$$

$$= \frac{1}{2n} \frac{\pi}{\sin\left(\frac{s\pi}{2n} + \frac{\pi}{2}\right)}$$
$$\overline{\left(\frac{\pi}{\sin\left(\frac{s}{n}+1\right)\pi}\right) \frac{1}{\sqrt{\frac{s}{n}}}}$$

$$= \frac{1}{2n} \frac{\cancel{\pi}}{\left(\cos\frac{s\pi}{2n}\right)} \times \left\{ \frac{\left(\sqrt{\frac{s}{n}}\right)}{\cancel{\pi}} \left\{ \sin\left(\frac{s}{n}\pi + \pi\right) \right\} \right\}$$

$\sin(A+B)$

$$= \frac{\sqrt{\frac{s}{n}} \; (-) \; \sin\left(\frac{s\pi}{n}\right)}{2n \cos\left(\frac{s\pi}{2n}\right)}$$

$$= \sqrt{\frac{s}{n}} \cdot \frac{\cancel{2} \sin\left(\frac{s\pi}{2n}\right) \cancel{\cos\left(\frac{s\pi}{2n}\right)}}{\cancel{2}n \cancel{\cos\left(\frac{s\pi}{2n}\right)}}$$

$$= \frac{1}{n} \sqrt{\frac{s}{n}} \quad \sin\left(\frac{s\pi}{2n}\right)$$

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Or $$\boxed{\int_0^\infty x^{s-1} \sin(x^n)\, dx = \frac{1}{n} \Gamma\left(\frac{s}{n}\right) \sin\left(\frac{\pi s}{2n}\right)}$$

$\forall s \in (-1, 1) n$

Fascinating Integral

let $s = 1$

$$\int_0^\infty \sin(x^n)\, dx = \frac{1}{n} \Gamma\left(\frac{1}{n}\right) \sin\left(\frac{\pi}{2n}\right)$$

$n = 2$

$$\int_0^\infty \sin(x^2) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \sin\left(\frac{\pi}{4}\right) = \frac{1}{2} \frac{\sqrt{\pi}}{\sqrt{2}} = \frac{1}{2}\sqrt{\frac{\pi}{2}}$$

We can use

* $\boxed{\sin x^n = e^{ix^n}}$ Imag part

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7) $f(x) = \cos(x^n)$

$$\cos(x^n) = \sum_{k=0}^{\infty} (-1)^k \frac{(x^n)^{2k}}{2k!}$$

$\rightarrow \sum (-1)^k (x^n)^{2k}$ $\cos\frac{\pi}{2}$ $k!$

$$I_n(s) = \int_0^{\infty} x^{s-1} \cos(x^n)\, dx$$

$$= \int_0^{\infty} x^{s-1} \sum (-1)^k \frac{(x^n)^{2k}}{2k!}$$

let $x^n = t$, $x = (t)^{1/n}$, $dx = \frac{1}{n}(t)^{\frac{1}{n}-1} dt$

$$\frac{1}{n}\int_0^{\infty} (t)^{\frac{s-1}{n}} \left(\sum (-1)^k \frac{(t)^{2k}}{2k!}\right) \cdot (t)^{\frac{1}{n}-1} dt$$

$$= \frac{1}{n}\int_0^{\infty} (t)^{\frac{s}{n}-1} \sum (-1)^k \frac{(t)^{2k}}{2k!}\, dt$$

let $t = \sqrt{l}$, $dt = \frac{1}{2\sqrt{l}}\, dl$

$$= \frac{1}{2n}\int_0^{\infty} (l)^{\frac{s-\cancel{n}}{2n}-\frac{1}{2}} \left(\sum (-1)^k \frac{(l)^k}{2k!}\, dl\right) \quad l^{-1/2}$$

$$= \frac{1}{2n}\int_0^{\infty} l^{\frac{s-1}{2n}} \sum \frac{(-l)^k}{2k!}\, dl$$

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$$= \frac{1}{2n}\sum_{k=0}^{\infty}\int_0^{\infty} t^{\frac{s}{2n}-1}\,\frac{(-t)^k}{k!}\cdot\underbrace{\left(\frac{k!}{2(k)}\right)}_{\lambda(-k)}\,dt$$

$$= \frac{1}{2n}\,\Gamma\frac{s}{2n}\quad\lambda\left(-\frac{s}{2n}\right)$$

$$= \frac{\frac{1}{2n}\Gamma\frac{s}{2n}\qquad\Gamma\left(1-\frac{s}{2n}\right)}{\Gamma\left(1-\frac{s}{n}\right)}$$

$$\left\{\because\ \Gamma n\,\Gamma 1-n = \frac{\pi}{\sin n\pi}\right.$$

$$= \frac{\left(\frac{1}{2n}\right)\left(\cancel{\sin t}\ \frac{\pi}{\sin\left(\frac{s}{2n}\pi\right)}\right)}{\Gamma\left(1-\frac{s}{n}\right)}$$

$$\left[\frac{1\quad\cancel{\pi}}{(2n)\left\{\sin\left(\frac{s\pi}{2n}\right)\right.}\right]\cdot\frac{1}{\left(\frac{\cancel{\pi}}{\sin\left(\frac{s\pi}{n}\right)}\right)}\cdot\frac{1}{\Gamma\frac{s}{n}}$$

$$= \frac{1}{\cancel{2}n\ \cancel{\sin\left(\frac{\pi s}{2n}\right)}}\cdot\frac{\cancel{2}\cancel{\sin\left(\frac{s\pi}{2n}\right)}\cos\left(\frac{s\pi}{2n}\right)}{\Gamma\frac{s}{n}}$$

$$= \left(\frac{1}{n}\right)\Gamma\frac{s}{n}\quad\cos\left(\frac{s\pi}{2n}\right)$$

$$\boxed{\text{or}\quad\int_0^{\infty} x^{s-1}\cos(x^n)\,dx = \frac{1}{n}\Gamma\frac{s}{n}\cos\left(\frac{s\pi}{2n}\right)}$$

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let $s=1$

$$\int_0^\infty \cos(x^n)\,dx = \frac{1}{n}\Gamma\frac{1}{n}\cos\left(\frac{\pi}{2n}\right)$$

$n=2$

$$\int_0^\infty \cos(x^2)\,dx = \frac{1}{2}\Gamma\frac{1}{2}\cos\left(\frac{\pi}{4}\right) = \frac{1}{2}\sqrt{\frac{\pi}{2}}$$

Ex-1) $$\int_0^\infty \frac{x^{1/a}}{x^2+1}\,dx$$

$$= \int_0^\infty x^{1/a}\sum_{n=0}^\infty \frac{\cancel{\Gamma n+1}}{\Gamma_1}\frac{(-1)^n (x^2)^n}{\cancel{n!}}\,dx$$

let $x^2 = t$

$$= \int_0^\infty (t)^{\frac{1}{2a}}\sum_{n=0}^\infty (-1)^n (t)^n \cdot \frac{1}{2\sqrt{t}}\,dt$$

$$= \frac{1}{2}\int_0^\infty (t)^{\frac{1}{2a}-\frac{1}{2}}\sum (-1)^n (t)^n\,dt$$

$$= \frac{1}{2}\sum\int_0^\infty (t)^{\frac{1}{2a}-\frac{1}{2}}\sum\frac{(-t)^n}{n!}\underbrace{n!}_{\lambda(n)}\,dt$$

$$= \frac{1}{2}\Gamma\frac{1}{2a}+\frac{1}{2}\ \lambda\left\{-\left(\frac{1}{2a}+\frac{1}{2}\right)\right\}$$

$$= \frac{1}{2}\frac{\Gamma\frac{1}{2a}+\frac{1}{2}}{\Gamma n}\ \frac{\Gamma\frac{1}{2}-\left(\frac{1}{2a}\right)+1}{\Gamma 1-n} = \frac{1}{2}\frac{\pi}{\sin\left(\frac{1}{2a}+\frac{1}{2}\right)\pi}$$

$$= \frac{\pi}{2}\left\{\frac{1}{\cos\frac{\pi}{2a}}\right\} = \boxed{\frac{\pi}{2}\sec\left(\frac{\pi}{2a}\right)}$$

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Ex.2) $\int_0^\infty \frac{x \log x}{(1+x^2)^2} dx = 0$

Hint: set $x = \frac{1}{y}$ to get $2I = 0$

let Solve

$$I = \int_0^\infty \frac{x^a}{(1+x^2)^2} dx$$

where $I'(I)$ = is our required integration

$$I = \int_0^\infty \frac{x^a}{(1+x^2)^2} = \int_0^\infty x^a \sum_{n=0}^{\infty} \frac{\Gamma(n+2)}{\Gamma(2)} \frac{(-1)^n (x^2)^n}{n!}$$

let $x^2 = t$, $x = \sqrt{t}$, $dx = \frac{1}{2\sqrt{t}}$

$$I = \frac{1}{2}\int_0^\infty (t)^{\frac{a}{2}-\frac{1}{2}} \sum \underbrace{\Gamma(n+2)}_{\lambda(n)} \frac{(-t)^n}{n!}$$

using RMT

$$= \frac{1}{2}\Gamma\left(\frac{a}{2}+\frac{1}{2}\right) \lambda\left\{-\left(\frac{a}{2}+\frac{1}{2}\right)\right\} = \frac{1}{2}\Gamma\left(\frac{a}{2}+\frac{1}{2}\right)\Gamma\left(\frac{3}{2}-\frac{a}{2}\right)$$

$$= \frac{1}{2}\left(\frac{1}{2}-\frac{a}{2}\right)\Gamma\left(\frac{a}{2}+\frac{1}{2}\right)\Gamma\left(1-\left(\frac{a}{2}+\frac{1}{2}\right)\right) = \left(\frac{1-a}{4}\right)\frac{\pi}{\sin\left(\frac{a}{2}+\frac{1}{2}\right)\pi}$$

$$I = \frac{1-a}{4} \frac{\pi}{\cos\frac{a}{2}\pi}$$

$$I' = \left(\frac{\pi}{\cos\frac{a}{2}\pi}\right)\left(-\frac{1}{4}\right) + \pi\left(\frac{1-a}{4}\right)\left(\sec\frac{a}{2}\pi \tan\frac{a}{2}\pi\right)\frac{\pi}{2}$$

$I'(1) = 0 \quad \lim a \to 1$

When ever seen log term, use $\frac{d}{dx} a^x = \underline{a^x \log a}$

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Ex) $\int_0^\infty \log^m x \{\sin(x^n)\} dx$

let

$$I = \int_0^\infty x^\phi \sin(x^n) dx$$

When $\frac{d^m}{d\phi^m} \int_0^\infty x^\phi \sin(x^n) dx$ is our integral.

$$\int_0^\infty x^\phi \sin(x^n) dx = \int_0^\infty x^\phi \sum_{k=0}^\infty (-1)^k \frac{x^{n(2k+1)}}{(2k+1)!}$$

$$= \int_0^\infty x^{\phi+n} \sum (-1)^k \frac{x^{2nk}}{(2k+1)!}$$

let $x^{2n} = t$, $x = (t)^{1/2n}$, $dx = \frac{1}{2n} (t)^{\frac{1}{2n}-1} dt$

$$\int_0^\infty (t)^{\frac{\phi+n}{2n}} \sum \frac{1}{(2k+1)!} \frac{(-t)^k}{k!} (k!) \cdot \frac{1}{2n} (t)^{\frac{1}{2n}-1} dt$$

$$\frac{1}{2n} \int_0^\infty (t)^{\frac{\phi+n+1}{2n}-1} \sum \frac{k!}{\underbrace{(2k+1)!}_{\lambda(k)}} dt$$

Using RMT

$$\frac{1}{2n} \Gamma\left(\frac{\phi+n+1}{2n}\right) \lambda\left(-\frac{\phi+n+1}{2n}\right)$$

$$= \frac{1}{2n} \frac{\Gamma\left(\frac{\phi+n+1}{2n}\right) \Gamma\left(1-\left(\frac{\phi+n+1}{2n}\right)\right)}{\Gamma\left(2-\frac{\phi+n+1}{n}\right)} =$$

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$$= \left(\frac{1}{2n} \frac{\pi}{\sin\left(\frac{\theta+n+1}{2n}\right)\pi}\right) \frac{1}{\sqrt{1-\frac{\theta+1}{n}}}$$

$$= \left(\frac{1}{2n}\right) \frac{\cancel{\pi}}{\cos\left(\frac{\theta+1}{2n}\right)\pi} \cdot \frac{\sin\left(\frac{\theta+1}{n}\right)\pi}{\cancel{\pi}} \sqrt{\frac{\theta+1}{n}}$$

$$= \frac{1}{\cancel{2}n} \sqrt{\frac{\theta+1}{n}} \frac{\cancel{2} \sin\left(\frac{\theta+1}{2n}\right)\pi \cancel{\cos\left(\frac{\theta+1}{2n}\right)\pi}}{\cancel{\cos\left(\frac{\theta+1}{2n}\right)\pi}}$$

$$= \frac{1}{n} \sqrt{\frac{\theta+1}{n}} \sin\left(\frac{\theta+1}{2n}\right)\pi$$

Hence

$$\int_0^\infty \log^m x \sin(x^n)\,dx$$

$$= \lim_{\theta\to 0} \frac{d^m}{d\theta^m} \left[\frac{1}{n} \sqrt{\frac{\theta+1}{n}} \sin\left(\frac{\theta+1}{2n}\right)\pi\right]$$

Use Complex Notation